

Bank of India



FortiFi

Intelligent Fraud Prevention & Adaptive Authentication for Mobile Banking

Behavior-Aware | Passwordless | Real-Time Detection

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Problem Statement

The Problem

- 75%+ of banking frauds stem from phishing, SIM swaps, and impersonated registrations
- OTPs & passwords are no longer secure entry points
- Fraudsters exploit app installations, device spoofing, and SIM hijacks
- Existing authentication is static, reactive, and exploitable

What are we solving?

- **Impersonated Registrations**
Fraudster registers using customer's number & stolen OTPs
- **SIM Swap & Device Hijack**
Remote attackers gain control over SIM or mimic device fingerprint
- **Lack of Adaptive Intelligence**
Rule-based systems fail to dynamically adapt to evolving behavior

Outcome Expected



Distinguish legit customers from fraudsters



Alert and prevent unauthorized access



Eliminate reliance on OTP/passwords

How We're Different from the Traditional Way

FortiFi addresses the problems of impersonated registrations, SIM/device hijacks, and lack of adaptive intelligence by combining:



Device + SIM intelligence



Behavioral biometrics during login



ML-based real-time fraud risk scoring



Adaptive authentication layers



Instant alerts & admin dashboard for fraud visibility

What Our Solution Does?

The Problem

Impersonated Registrations

SIM Swap / Device Hijack

Lack of Adaptive Intelligence



FortiFi Solution

Captures SIM ID, device ID, IP, geo-location & behavior patterns during registration

Detects anomalies in SIM/device fingerprint vs. history

Uses ML model to dynamically score fraud risk and recommend action (login, verify, block)



Outcome

Secure logins, zero OTP reliance, early fraud detection, without disrupting legit users.

Technical Deepdive – Architecture

Login Data Collector

- Captures: Device ID, SIM info, IP, location, touch/typing patterns
- Input: Raw login metadata



Feature Engineering Layer

- Converts raw login data into structured features
- e.g. SIM consistency, device fingerprint match, typing anomaly score



ML-Based Risk Scoring Engine

- Model: XGBoost / Random Forest
- Combines all features → outputs risk score (0–10)



Adaptive Authentication Layer

- Low Risk → allow login
- Medium Risk → Face ID or KYC prompt
- High Risk → block + trigger fraud alert



Alerts & Admin Dashboard

- Logs all suspicious activities
- Exposes risk score breakdown (for explainability)
- Allows manual review of blocked cases



Implementation Stack & Strategy

Data Simulation

- Using public fraud datasets for reference
 - <https://www.kaggle.com/datasets/sriharshaeedala/financial-fraud-detection-dataset>
 - <https://www.kaggle.com/datasets/valakhorasani/bank-transaction-dataset-for-fraud-detection>
- Generated synthetic behavioral data using GANs (Generative Adversarial Networks) to simulate diverse user login styles and to create edge-case fraud scenarios (e.g., bots mimicking typing)

Tech Stack



Frontend UI



Flask



python



Backend



ML Engine



Data Pipeline

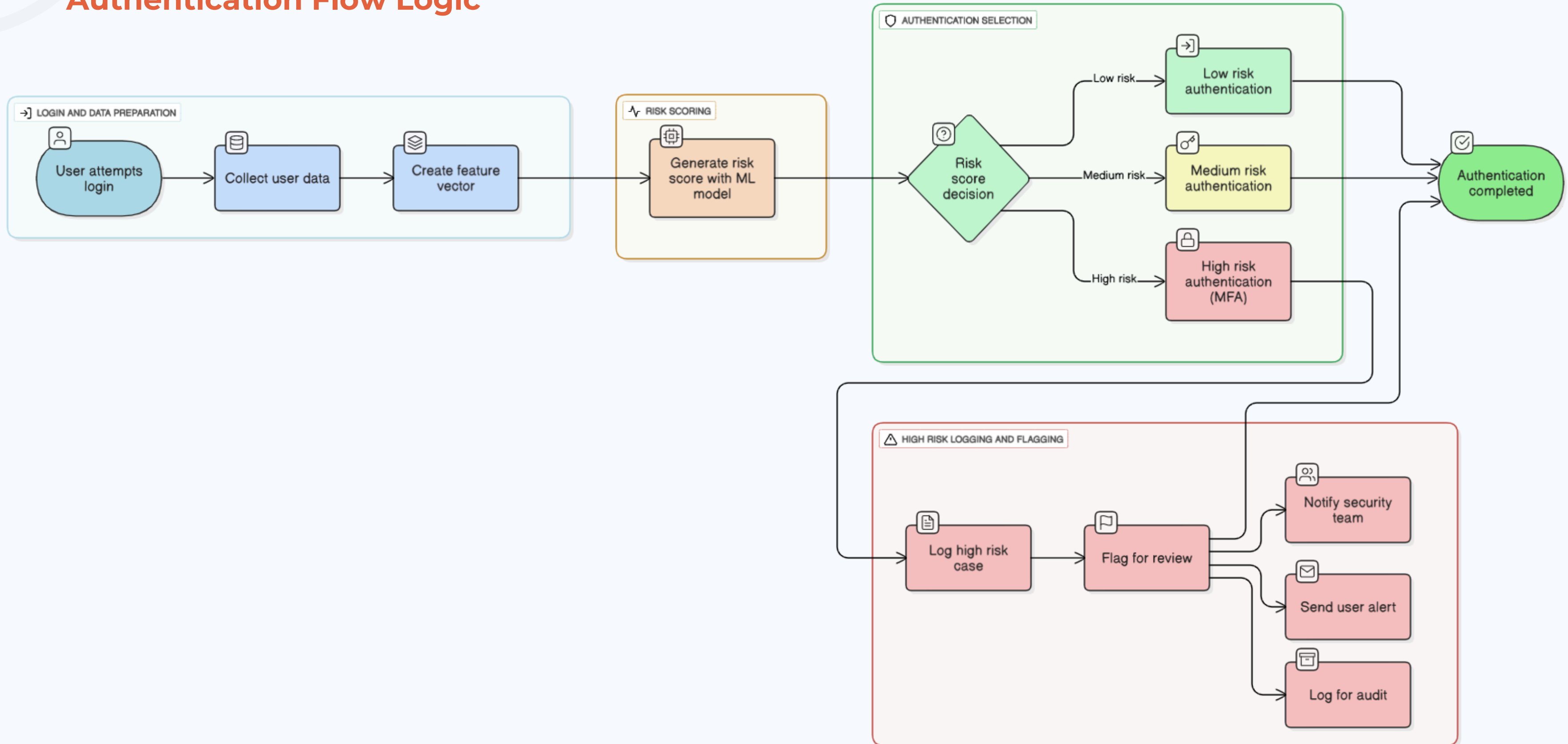


Streamlit

Dashboard

Implementation Stack & Strategy

Authentication Flow Logic

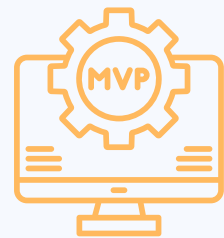


Feasibility of Our Solution



Technically Feasible

- All components (device fingerprinting, ML risk scoring, typing biometrics) are possible using existing tools/APIs
- Synthetic dataset and emulator-based testing works without real user data



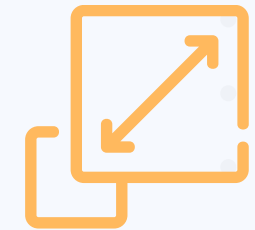
Business Viable

- No friction for legit users → improves UX
- Blocks fraud before financial transaction happens
- Scalable to any bank/fintech app with SDK/API integration



Regulatory Friendly

- Does not rely on sensitive financial or personal documents
- Can be made compliant with DPDP/GDPR (behavioral data stays device-side if needed)



Extendable

- Future integration: voice biometrics, image-based spoof detection
- Can support onboarding (not just login) for full fraud-free KYC

THANK YOU

